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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 02/11/2022 |
| **PI Name:** | Hyde, Truell Ph.D. | | |
| **Project Title:** | PK-4: Self-Ordering of Interacting Complex Plasma Particles in Microgravity | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Fundamental physics | | |
| **Joint Agency Name:** | National Science Foundation | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS |
| **Start Date:** | 04/19/2017 | **End Date:** | 12/31/2023 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 7 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | Callas, John | **Contact Phone:** | |
| **Contact Email:** | john.l.callas@jpl.nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 12/31/2023 per U. Israelsson/JPL (Ed., 1/6/22)<br>NOTE: End date changed to 8/31/2022 per U. Israelsson/JPL (Ed., 8/12/21) | | |
| **Key Personnel Changes/Previous PI:** | None. | | |
| **COI Name (Institution):** | Matthews, Lorin Ph.D. ( Baylor University )<br>Hartmann, Peter Ph.D. ( Baylor University )<br>Petrov, Oleg Ph.D. ( Joint Institute for High Temperatures, Russian Academy of Sciences )<br>Rosenberg, Marlene Ph.D. ( University of California, San Diego ) | | |
| **Grant/Contract No.:** | JPL 1571701 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

**Task Description:**

Abstract --
Complex plasma consists of ions, electrons, neutral gas, and solid microparticles referred to as dust. Dust grains immersed in plasma become negatively charged and interact with one another as well as plasma components through electromagnetic interactions. In particular, dust grains experience a plasma drag force, altering the trajectories of ions in the plasma, and participate in collective plasma interactions, such as waves and instabilities. An interesting aspect of complex plasma is its ability to self-organize into dust liquids, 2D and 3D dust crystal lattices, and 1D dust chains, which are aligned with electric fields in the plasma. This feature provides researchers the ability to examine classical collective phenomena on size and time scales allowing for direct observation (at the kinetic level) of fundamental microscopic and macroscopic processes, as the dust can easily be imaged with high-speed cameras.

In many environments, positive ions in the plasma have a directed flow with respect to the negatively charged dust grains. The resulting interaction between the dust and flowing plasma creates an ion wakefield downstream from the dust particles, with the positive space region modifying the interaction between the charged grains and contributing to the observed dynamics and particle structure. Since the ion wakefield is weak, these effects are often masked by gravity for terrestrial experiments. This international research collaboration employs the Plasma Krystall-4 (PK-4) on the International Space Station (ISS) to conduct experiments in microgravity which investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains and other ordered structures.

Scientific Goals / Objectives of the Research --

The goal of the proposed research project is to investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains in a microgravity environment. This goal will be accomplished by pursuing the following specific objectives: 1. Establishing the complex plasma parameters required for the formation and stability of extended field-aligned dust particle chains, 2. Measuring the grain-grain potential along and perpendicular to the ion flow direction, and 3. Studying interchain interactions and their role in the structural properties of an ordered complex plasma system.

Significance of the Research --

This research will be used to develop comprehensive theories on interparticle potential and grain-plasma interactions which determine the microscopic and macroscopic properties of complex plasma, including external forces acting on the particles, the fundamental interactions between the particles, thermodynamics, and self-organization of the system. The results will directly impact research in astrophysics and planetary science (dust in protoplanetary disks, early protoplanetary development, charged dust on the surface of airless bodies, such as Mars, the moon, and asteroids; planetary ring systems), atmospheric physics (noctilucent clouds), fusion research (dust contamination, operating stability), materials physics (soft matter, metastable states, complex fluids, granular matter), and advanced manufacturing (semiconductor, etching, and deposition systems).

**Rationale for HRP Directed Research:**

Impact on Principle Discipline –
Experimental (Baylor & Wigner Research Centre for Physics, Budapest) – Over the past five years, the research analog to the PK-4 (ISS), the PK-4 BU, has been constructed, calibrated, and commissioned. The PK-4 BU is currently the only device of this sort in the world which allows researchers to identify new parameter regimes or develop experimental plans which cannot be developed (or tested) on current PK-4 ground units. Specifically, during the initial two years of operation (2019 - 2020), the PK-4 BU was calibrated against data collected by both the PK-4 ISS and the PK-4 ground unit at DLR (German Aerospace Center) Munich. Additionally, the PK-4 BU was calibrated against the initial ground test data collected during the construction of the PK-4 ISS (before launch) to verify that both PK-4 BU Langmuir probe data and LIF (logical interface) data were in agreement with the final calibration data produced before the PK-4 ISS was installed on board the ISS. In both cases this data was collected with the help of our collaborators, with the LIF data collected in collaboration with Caltech and the Langmuir probe data collected in calibration with the Institute of Space Systems (IRS) at the University of Stuttgart.

The above has resulted in the PK-4 BU now being employed successfully for identification of appropriate parameter spaces for experiments aboard the space station and for assistance in the analysis of previous PK-4 ISS data showing anomalies. Specifically, this process has been used by researchers from Baylor, Auburn, and DLR to establish parameter spaces for the proposal of multiple experimental campaigns employing the PK-4 ISS. To date, Flight Campaigns 7, 9, 12 and 14 have either been accepted or flown on board the ISS. As of January 2021, the data from Campaign 7 is under analysis with the first of several papers soon to be submitted. Unfortunately due to the global pandemic, data from Campaigns 9 and 12 has not yet been delivered to the researchers. Campaign 14 is slated for flight in 2022. Hopefully, this situation will resolve itself before funding runs out in 2024.

Numerical (Baylor) – Along with the development of the PK-4 BU, an entirely new code, DRIAD (Dynamic Response of Ions and Dust) has been written to model the ion-dust interactions within the PK-4 BU discharge. This allows both dust charging and dynamics to be self-consistently calculated, including the effects of the ion wakefield on the dust-dust interactions and charging. The code is written in C++ with CUDA to make use of the parallel processing power of Graphical Processing Units (GPUs). The code makes use of the boundary conditions for the plasma provided by a PIC-MCC code (developed by our collaborators at the Wigner Research Centre for Physics in Hungary) used to model the PK-4 DC discharge. DRIAD is a MD simulation of ion and dust dynamics, each resolved on their own timescale. The charging of the dust is calculated self-consistently within the ion wake. The dynamics of the ions include calculations of ion-ion accelerations, ion-dust accelerations, ion-neutral collisions, ion insertions (assuming a shifted Maxwellian velocity distribution incident on the simulation boundary), and ion deletions due to leaving the simulation boundary or impacting the dust particle. Dust dynamics include calculations of charge, ion-dust accelerations, dust-dust accelerations, external electric fields present in the DC plasma, additional confining electric fields, gas drag, and a thermal gas bath.

The code has been tested using boundary conditions provided by the aforementioned PIC-MCC code, showing the effects of the plasma striations (which occur on the microsecond timescale) on the dust dynamics (which occur on the millisecond time scale).

While many features implemented within the code are specific to the PK4 discharge, the code is robust and can be

**Research Impact/Earth Benefits:**

applied to other dusty plasma experiments by changing the boundary conditions governing the ion flow and external confining potentials. This allows users of the code to explore the parameter space of experimental conditions to determine which are most favorable for a particular dust configuration, such as one-dimensional strings. A second version of the code is under development which takes advantage of capabilities in C++ to allow rapid implementation of new forces, boundary conditions, elements, etc. This version of the code will also be adaptable beyond dusty plasma physics.

Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) -- The results of the PIC/MCC simulations of the PK-4 gas discharge predicting the presence of fast ionization waves during both stationary DC discharge and during polarity switching have now been experimentally verified at Baylor employing the PK-4 (BU). These rapid features remained undetected in the PK-4 (ISS) experiment due to lack of fast diagnostics but are apparently of key importance for the formation of the observed dust particle chains. (See Hartmann, P., M. Rosenberg, Z. Juhasz, L. Matthews, D. Sanford, K. Vermillion, J. Reyes, T. Hyde, Ionization waves in the PK-4 direct current neon discharge, Plasma Sources Science and Technology, 29(11), 115014, November 2020.) These findings provide the necessary foundation for design of future experimental campaigns.

Numerical (USCD) –The UCSD PI has been developing theory for longitudinal and transverse dust acoustic waves in a one-dimensional chain of dust particles in the strongly coupled liquid phase. There is a harmonic confining potential perpendicular to the chain. The formulation uses the quasi-localized charge approximation (QLCA) which requires input of the pair correlation function of the dust. Collisions of dust with neutrals are included. Owing to the high collisionality of PK-4 experiments (e.g., at pressures of 40-60 Pa), it appears likely that longitudinal waves would be damped. However, transverse waves, whose dispersion depends also on the magnitude of the confining potential, typically have higher frequencies and may propagate. If transverse waves are observed in PK-4 experiments, this may provide information on the magnitude of the perpendicular confining force. The theoretical analysis needs to be further developed using pair correlation function data from experiments and/or simulations, and dust interaction potentials that may be derived from simulations.

Impact on Scientific Goals / Objectives of the Research

Objective 1: Establishing the plasma parameters required for the formation and stability of dust particle chains.

Experimentally, the PK-4 BU has now been employed successfully for identification of experimental parameter spaces for both current and future experiments aboard the space station and analysis of previous PK-4 ISS data showing anomalies. This process culminated in the summer of 2019 with the PK-4 BU being used by researchers from Baylor, Auburn, and DLR to establish specific parameter spaces for two experimental campaigns on the PK-4 ISS. These campaigns (Campaign 7 and Campaign 9) were accepted for flight and flown on the PK-4 ISS in November of 2019 and February of 2020.

Additional Flight Campaigns 12 and 14 have either been accepted or flown on board the ISS. As of January 2021, the data from Campaign 7 is under analysis with the first of several papers which will come out of this data just published. Unfortunately due to the global pandemic, data from Campaigns 9 and 12 has not yet been delivered to the researchers. Campaign 14 is slated for flight in 2022. Hopefully, this will situation resolve itself before funding runs out in April of 2024.

Numerically, the MD model has now been used to show that there is a difference in dust behavior when time-averaged values of the plasma parameters (i.e., electric field, ion and electron density and temperature) are used, compared to the evolving plasma parameters, even when these occur on a microsecond time scale. An important part of this is the extinction/reignition of the plasma during the DC polarity switching of the PK4 electrodes. The average ion concentration around the dust particles is found to be much stronger resulting in ‘deeper wakes’ in the electric potential when the time-evolution of the plasma conditions is resolved.

Objective 2: Measuring the grain-grain potential along and perpendicular to the ion flow direction.

Experimentally / Numerically, the grain-grain potential along and perpendicular to the ion flow direction has been shown to differ significantly from previous analytical models used to explain the formation of dust chains.

Objective 3: Studying interchain interactions and their role in the structural properties of an ordered complex plasma system.

Experimentally, the PK-4 BU has now been employed to identify and reproduce multiple parameter sets favorable to dust string formation and modification of the internal structural properties of these systems. Numerically, preliminary models have been run using up to 1000 dust grains to examine the interaction of multiple chains. One interesting discovery was that particles form nested cylinders arranged in a hexagonal mesh wrapped around each cylindrical surface. Axial cuts through these cylinders reveal ordered linear structures, or chains.

Significant Results

Using data collected experimentally on the PK4-BU using time-resolved boundary conditions provided by the PIC-MCC model, it has now been shown that the ionization waves observed experimentally in the DC column lead to large variations in particle charge on the microsecond time scale. The PIC-MCC model has also revealed that the plasma is completed extinguished and reignites over a time span of approximately 200 μs each DC switching cycle. This causes the dust to discharge and then recharge over the same time span. These large variations in the dust charge are most likely the main contributors to the anomalous heating observed for dust particles in experiments.

Impact on Other Disciplines

The field of complex plasmas is now considered the plasma state of Soft Matter. As such, the results reported here also directly impact research in this field.

Additionally, a second version of the code, RAPID (Rapid Accelerates Physics sImulation Development) is under development which takes advantage of capabilities in C++ to allow rapid implementation of new forces, boundary conditions, elements, etc. This version of the code will be adaptable beyond dusty plasma physics and be broadly applicable to many types of problems in physics.

Impact on the Development of Human Resources and Professional Development

CASPER (Baylor) – Over the past year, (April 2021 – January 2022) seven undergraduate students, seven graduate students and three Assistant Research Professors were directly involved in either the experimental portion or the numerical portion of the PK-4 project at Baylor. Four of the five undergraduate students mentioned are members of under-represented minorities in physics and three of the graduate students are female.

As part of this project, each of these students gained experience in either experimental or numerical skills as well as practice in analyzing data, writing scientific papers and presenting their results in international forums.

Additionally, experimental students also learned required safety skills (i.e., laser safety, basic operating skills in HV, lasers, vacuum systems and diagnostics) while numerical students gained experience in testing and developing code and programming using CUDA, C++ and Matlab. (Each of these programming languages is listed among the top twenty programming languages for 2017 by IEEE.) All students also gained experience in both technical writing and speaking.

In 2021/22, the PI's, graduate and undergraduate students presented their work in multiple presentations at the IEEE International Conference on Plasma Science (ICOPS), the Annual Meeting of the American Physical Society-Division of Plasma Physics (APS-DPP), the Annual Meeting of the Division for Planetary Sciences (American Astronomical Society) and the Gaseous Electronics Conference. Students also presented their results as part of the Baylor Physics Department colloquium series and made poster presentations during Universities Space Research Association (USRA) Scholars Week.

All students listed gained experience in both technical writing and speaking.

In addition, the PI presented research results at DPG, the Texas A&M, Baylor, Princeton Summer School on Quantum Science and Engineering (invited talk), the Lunar Surface Science Workshop: Fundamental and Applied Lunar Surface Research in Physical Sciences, the American Society for Gravitational and Space Research (ASGSR) Conference and participated in multiple NASA BPS Decadal Study Workshops.

Finally, over the past year CASPER students (both graduate and undergraduate) have had the opportunity to attend virtual meetings with the PI and our Co-Is from all over the world (Peter Hartmann, Hubertus Thomas, Vladimir Nosenko, Marlene Rosenberg).

Impact on Physical / Numerical Resources –

Experimental (CASPER, Baylor & Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) --

The PK-4 BU is a new tool that continues to prove itself useful for the design and testing of potential experiments employing the PK-4 ISS. The PK-4 BU is now being used by multiple US research groups (both faculty and graduate students) and should prove useful for years to come. Motivated by new insights provided by integration of numerical simulations of DC gas discharge plasma (Wigner Research Centre for Physics in Budapest, Hungary) with experimental data provided by the PK-4 BU device (the Hyde laboratory within CASPER at Baylor University) verification of fundamental calibration data has now been completed. This has proven invaluable to the identification, design, and planning of future experiments on the PK-4 facility ISS with four experimental campaigns (Campaign 7, 9, 12 and 14) now accepted for flight.

Numerical (Baylor) - The IonWake code and the DRIAD (Dynamic Response of Ions and Dust) code are both new tools which will prove useful for many years to come. In the PI’s experience, computer codes which provide new capabilities are used for ten to twenty years (or more) and provide a research platform for many future graduate and undergraduate students. As an example, previously developed codes have been used by about 20 undergraduate students and 3-5 graduate students over a period of ten years.

Impact on Society –

The impact of this project on graduate students, undergraduate students and K-12 students continues to increase the overall science literacy of the voting public, providing an impact beyond the bounds of science and engineering alone.

Collaborator Meetings –

Due to the COVID pandemic, the majority of collaborator meetings held over the past two years have been held on-line. Although not preferred, the ability to meet on a semi-regular basis multiple times during the year has proven to have some advantages.

New Technology: Installation and alignment of additional video equipment; LIF assessment of plasma across multiple regimes; Langmuir probes developed and installed with data compared to original source data.

TASK PROGRESS
Experimental (Baylor University) –

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the help of our collaborators, with the LIF data collected in collaboration with Caltech and the Langmuir probe data collected in calibration with the Institute of Space Systems (IRS) at the University of Stuttgart.

Numerical (Baylor / Wigner Research Centre for Physics, Budapest) –

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**Task Progress:**

Theoretical (University of California San Diego / UCSD) –

A. PK-4 discharge modeling. The UCSD Principal Investigator (PI) collaborated on a theoretical study of the PK-4 cylindrical discharge. The study was geared toward determining plasma parameters to be used for studies of dust dynamics in this device. The numerical investigation used a two-dimensional, electro- static, PIC/MCC discharge simulation code. The simulations showed that the positive column is inhomogeneous on short timescales with rapidly moving, high amplitude ionization waves dominating the structure. As a result, the local plasma parameters fluctuate at frequencies of the order of 10 KHz and with amplitudes up to 10 times the time-averaged values. The oscillatory behavior of the plasma environment is expected to affect the charging and chain formation of dust particles in PK-4 experiments. For example, the ion drift speed is a crucial parameter in the electrorheological model of chain formation, and the electric field of the ionization waves can significantly affect the ion drift speed. (See P. Hartmann, M. Rosenberg, Z. Juhasz, L. S. Matthews, D. L. Sanford, K. Vermillion, J. Carmona-Reyes and T. W. Hyde, 2020, Ionization waves in the PK-4 direct current neon discharge, Plasma Sources Sci. Technol. 29, 115014.)

B. Chain formation in PK-4. The UCSD PI has been collaborating with the Baylor and WRC groups on studies of how fluctuations in local plasma parameters due to ionization waves in PK-4 can affect the formation of chains of dust particles. Papers are in preparation.

C. Waves in electrorheological dusty plasmas. The UCSD PI worked on theoretical studies of waves in a one-dimensional dust chain in an electrorheological dusty plasma, where the dust particles interact via combined Yukawa and dipole-dipole potentials. The configuration of the dust particles is given by a pair correlation function, and the chain is confined in the perpendicular direction by a parabolic potential. We are using pair correlation functions and confinement potentials provided by simulations of chain formation by the Baylor group. Both longitudinal and transverse waves are being considered, and dust-neutral collisions are taken into account. Work is continuing.

D. Ion-dust streaming instability. Dust acoustic, or dust density, waves can be excited by the flow of ions via an ion-dust streaming instability. Dust acoustic, or dust density, waves have been observed in numerous ground-based experiments, and also in PK-4 experiments (e.g., Jaiswal et al., 2018, Phys. Plasmas 25, 083705; Goree et al., 2020, Phys. Plasmas 27, 123701). The UCSD PI, together with colleagues at UCSD, began to study the nonlinear development of the ion-dust streaming instability in parameter regimes where the ion and dust collision frequencies are a significant fraction of their respective plasma frequencies. This regime may be relevant to ground-based or microgravity experiments with high neutral pressure. The study used a one-dimensional particle-in-cell electrostatic simulation code that included thermalizing collisions with the background neutrals. Using a particular set of nominal parameters with an ion thermal Mach speed of 3, the simulations found cnodial-like dust density waveforms (i.e, waveforms with broad troughs and sharp crests) in the nonlinear phase. The study may have relevance for high pressure environments of PK-4, but further work is needed to model the ion-dust instability using representative PK-4 parameters. (See K. Quest, M. Rosenberg and A. Levine, Simulations of ion- dust streaming instability in a highly collisional plasma, 2020, J. Plasma Phys. 86, 905860605.)

E. Possible use of Mie-resonant microparticles. Recently there has been interest in the use of dielectric particles with high refractive index n in novel optical and photonic devices (e.g., Yang et al. 2017, Phys. Reports 701, 1-50). The UCSD PI considered theoretically the scattering of far-infrared radiation in a dusty plasma composed of spherical, high-n dielectric particles of radii ~ 4 - 5 μm. In particular, the scattering of radiation in the 30 - 40 μm range from aluminum oxide or mag- nesium oxide particles was studied. Using Mie theory, it was found that a large scattering efficiency can occur at magnetic dipole resonance wavelengths, and there can be directional scattering of electromagnetic waves (e.g., for a fixed particle size, the scattering can be pre- dominantly backward in certain wavelength intervals). Possible applications were considered, including partial mirror behavior, enhanced extinction, and particle size analysis. Possible complications were also considered, including issues related to particle heating which leads to constraints on the background gas pressure as well as the intensity of the incident radiation. A paper is in preparation: M. Rosenberg, “On scattering in a dusty plasma composed of Mie-resonant dielectric particles”.

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